

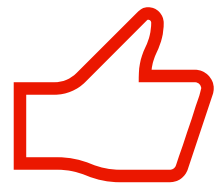
Final Co-op Presentation Shareable

August 13, 2025

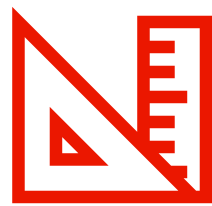
5480 Great America Parkway

Johnson & Johnson
MedTech

Agenda



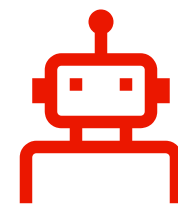
Introduction



PDU Redesign



Thermometer



Build-a-thon



Concluding
Remarks



Questions

Introduction

About Me

I'm a 4th year Electrical & Computer Engineering
Student @ Northeastern University

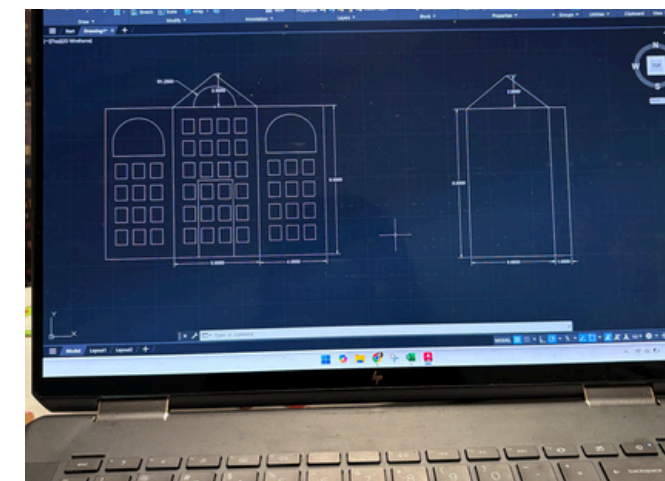
Hobbies: swimming, music, reading playing with my dog,
knitting, embroidery



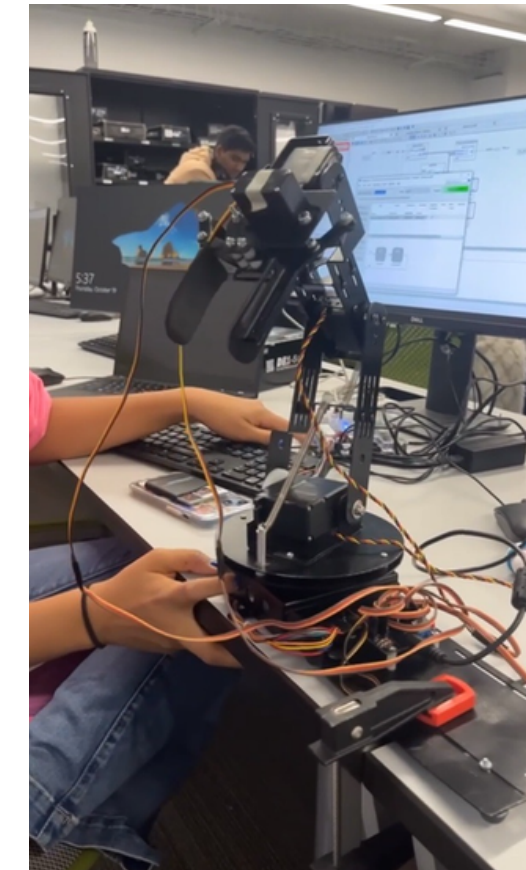
MY GOLDENDOODLE CHEETO!



RECENT EMBROIDERY



AUTOCAD DESIGNED
GINGERBREAD HOUSE

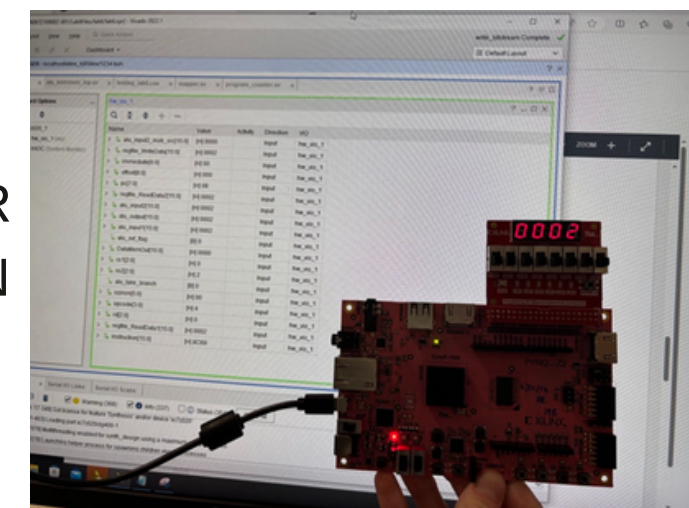


EMBEDDED ROBOTIC ARM



QUICK SORT SCARF
ROBOTIC KNITTING
RESEARCH LAB

RISCV-5
PROCESSOR
CREATED IN
SYSTEM
VERILOG



Power Distribution Unit Redesign

Power Distribution Unit

Goal: Redesign PDU for new needs

- Review Requirements & Specifications
- Address end of life components
- Find addressable MODEFs/CAPAs
- Remove Ether CAT Portal & GPIO Expanders
- Additional clean up
- Added functionality

Requirements & Specifications

- JAMA
- Reviewed requirements
- Compiled a list of requirements that may change based on relevance
- ~65 related SSS: may need to be updated based on prev redesign
 - EtherCAT, Power Control, Communication, Umbilical

Learnings from PDU

- Robotic PCB Design
 - Helped prepare for thermometer
- Power System & Controls Design
- PCB Troubleshooting
- Jama
- Specifications & Requirements
- Agile
- End of Life Components
- Design Lifecycle

Thermometer

Fluid Management System

- External company acquired company manufacturing the PCBs for fluid management
- Project goal was to transfer knowledge of the Board to Auris
- My role was to understand the electrical architecture and perform benchtop testing using the test fixtures

Uro Fluid Management System Test Board Documentation

7 PCBs in fluid system

- Main, Heater, Aspiration, Temperature, Pressure, IR Sensor, & USB

Challenges

- Unfamiliar test fixtures & missing documentation
- Inconsistent naming conventions
- Unknown signal connections

Benchtop Testing

- Tested the PCBs in the lab by following the documentation we received
- Verified expected and observed values
- Investigated root causes of any anomalies
- Identified missing tests and filled necessary gaps

Signal Tracing in Altium

- Cross referencing Altium Test Board & PCBs
- Learning how signals move through the PCB
- Tracing connections through layout & schematic
- Tracing signals between the pcb and test fixture to understand connections between the two and update the testing protocol

Confluence Documentation

- Step by step traced through the test procedure tests & mapped signals to Altium PCB & Schematic
- Explain each test & how the signals are traveling the board
- Included screenshots of the boards
 - Test board schematic
 - PCB Layout connections
 - PCB schematic connections to test board

Learnings from Thermometer

- Step by step tracing took a long time for each section of heater & main PCBs
- Connections between many PCBs: Main, Test Fixture, PDU
- Benchtop testing for test fixtures
- Understanding is one thing, explaining & documenting cohesively is another!
- Understanding the why behind the designs & learning to question them
- Altium!
- File paths & debugging

“You’ve got to dig a little deeper”

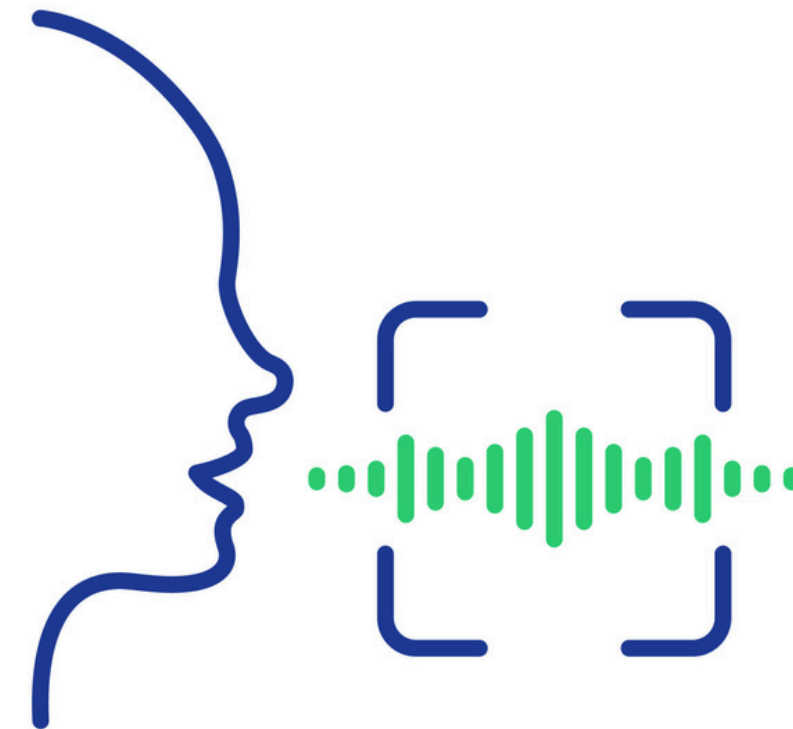
Build-A-Thon

Voice Control for Monarch

Thank you Andrea, Harsha,
Michael, Ning & Susan!

Why voice control?

- Hands-Free operation
- Increased Efficiency
- Improved Sterility
- Enhanced Productivity



Different Voice Control Modules

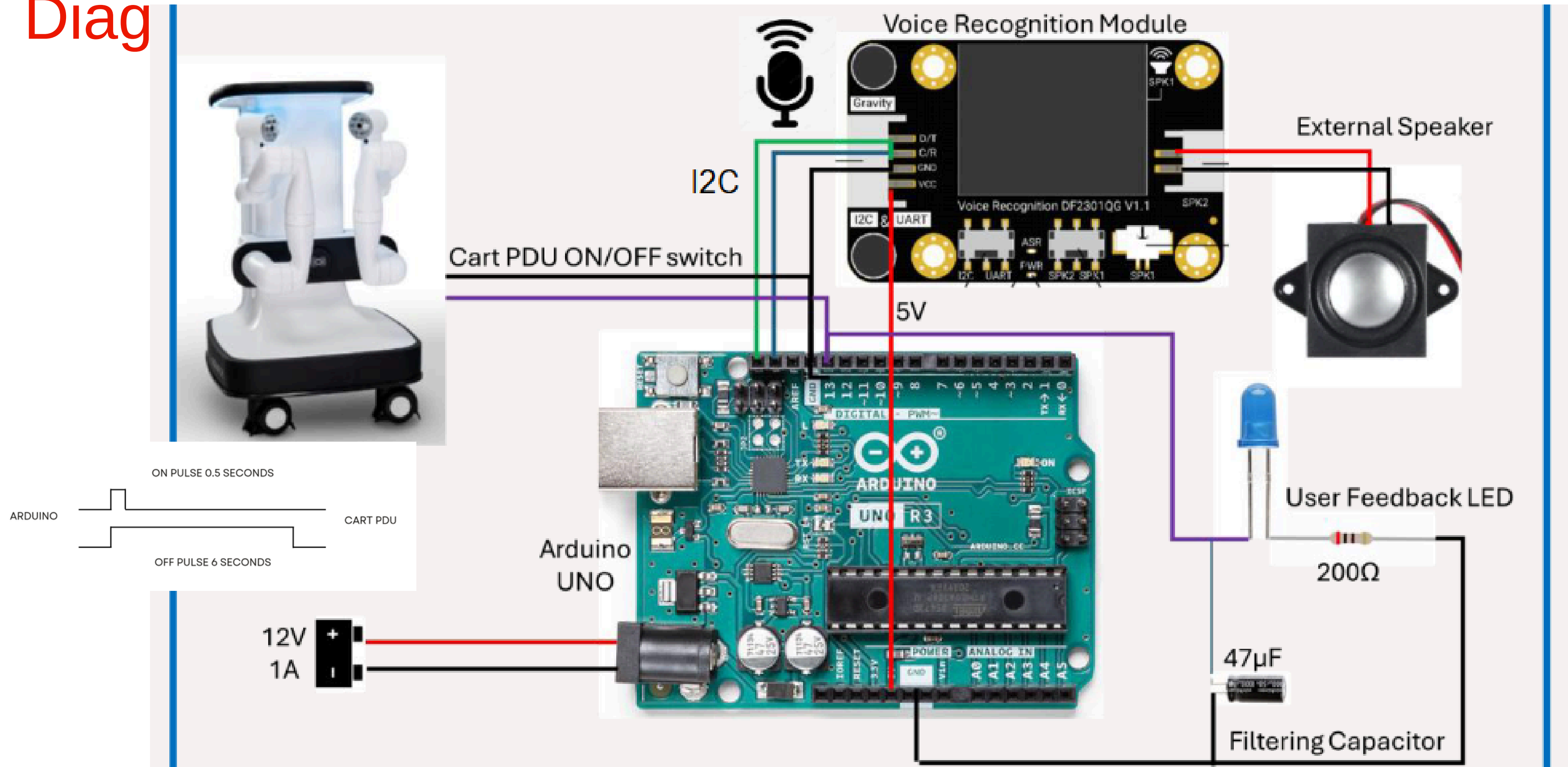
WWZMDiB Voice Recognition Module



DFRobot Voice Control Module

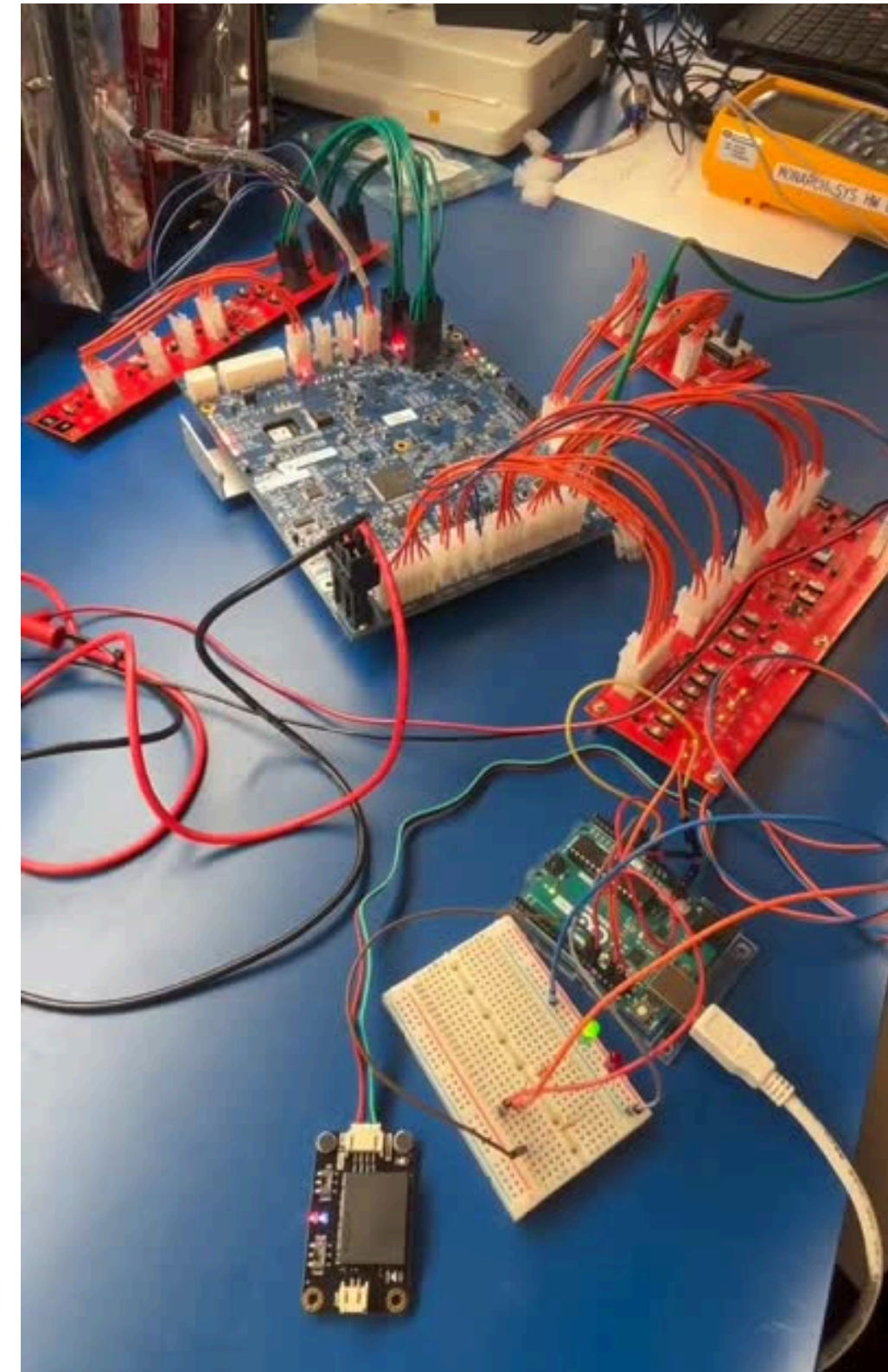
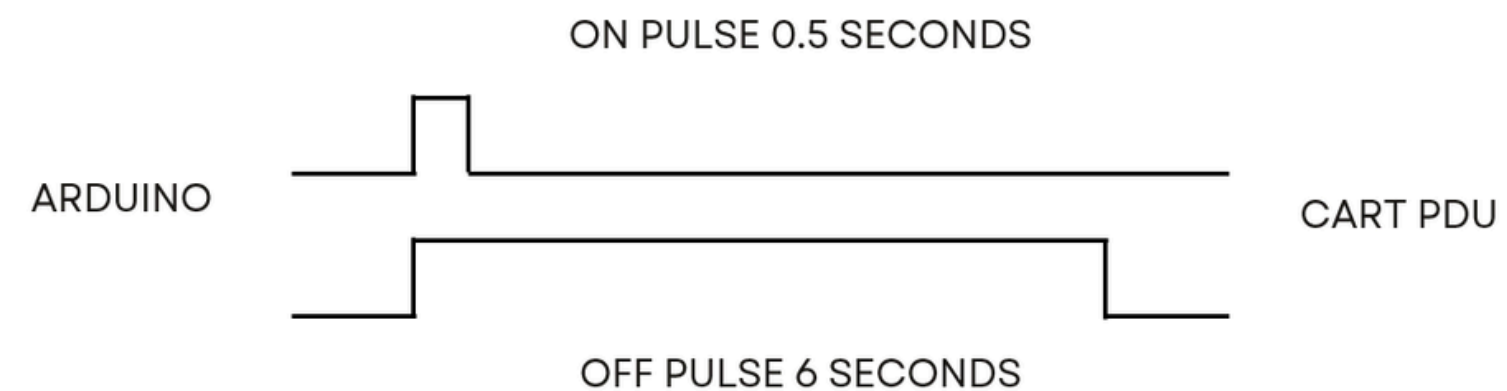


Electrical Hardware Block Diag



Benchtop Testing

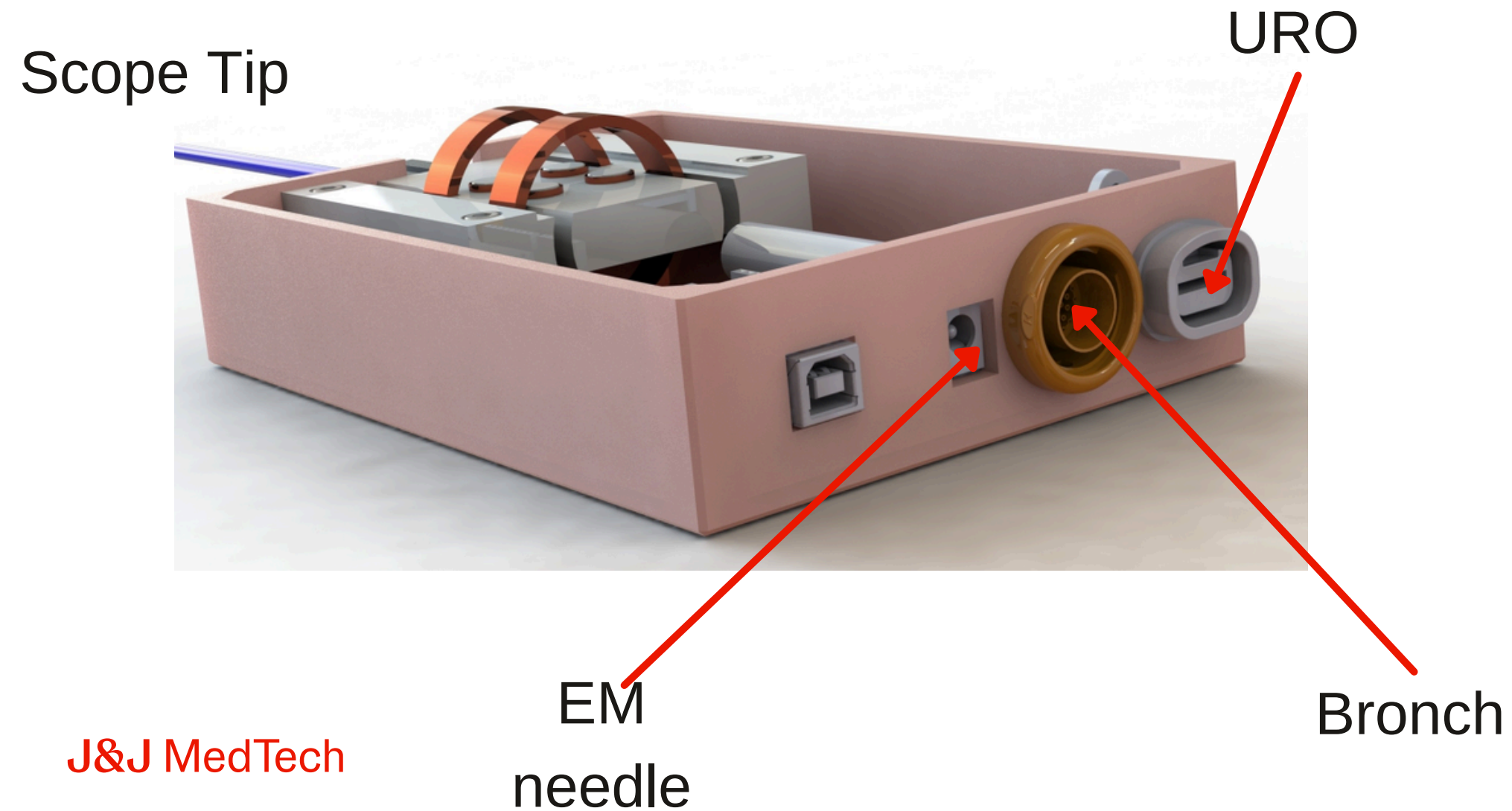
Used existing MOSFET on Cart PDU to control on & off signal



First time sending pulse to cart PDU to turn on & off!

EM Smart Tester

- Confirm EM sensor placement is correct during manufacturing



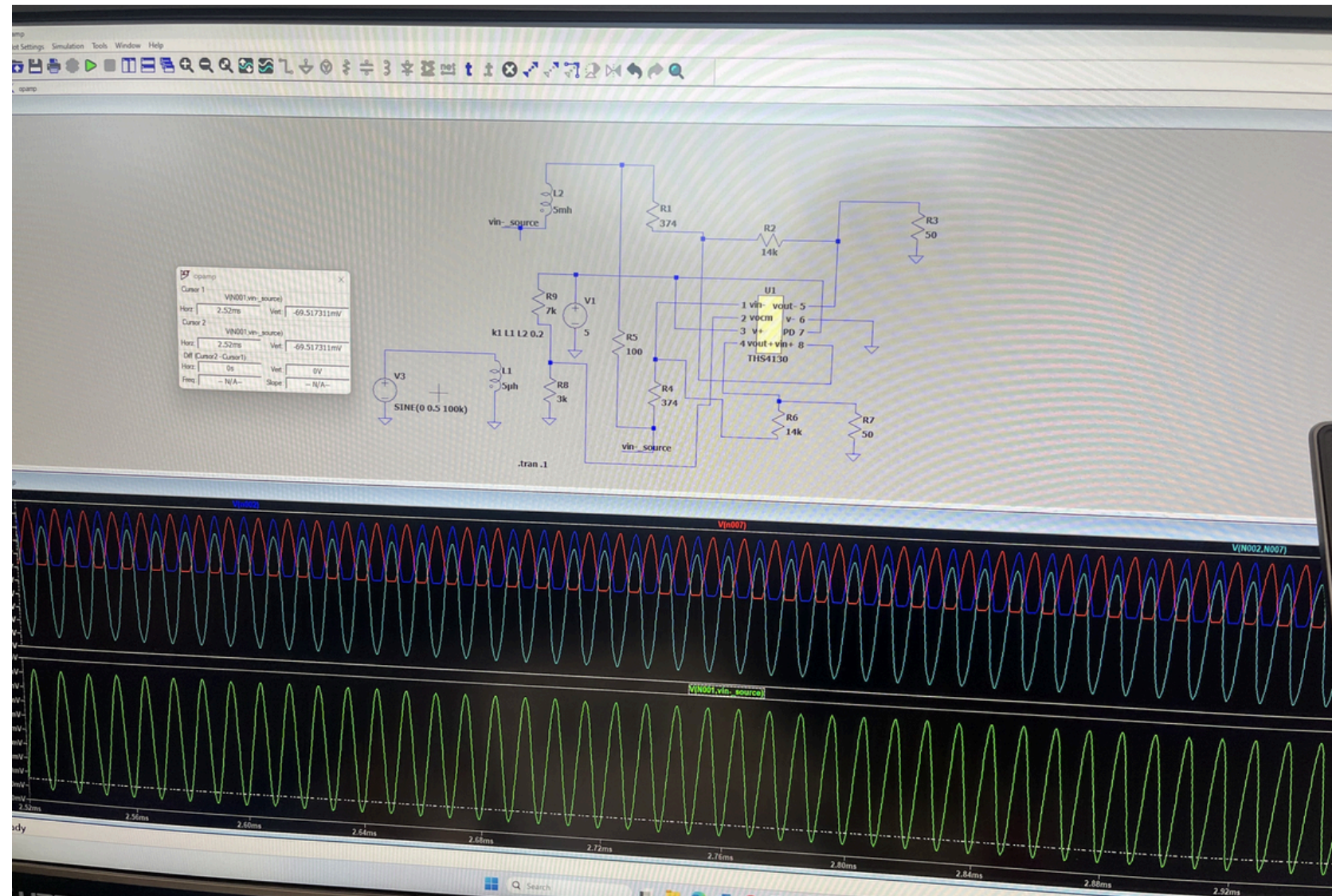
J&J MedTech



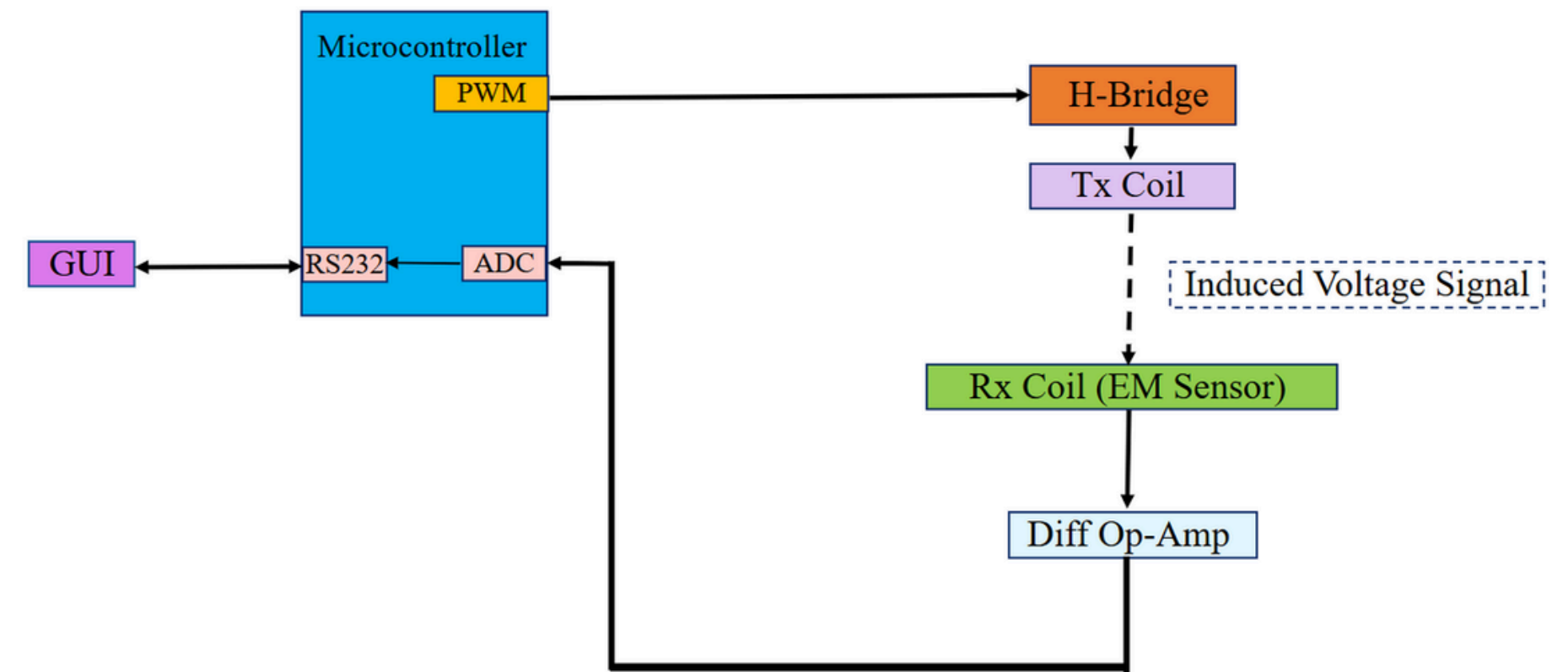
FIRST WORKING COIL VOLTAGE READS

Thank you Esther & Jason!

EM Smart Tester



LTSPICE SIMULATION



ELECTRICAL BLOCK DIAGRAM

Learnings from Build-A-Thon

- Off the shelf voice recognition module
- PDU understanding
- Integrating the module to the system
- Mechanical, electrical, and software integration
- EM coil winding & physics behind
- Debugging in the lab with differential op amp & driver
- Meet new people & work in teams!

“#NOSLEEP
TILDEMO”

Concluding Remarks

Learnings Beyond the Technical

- 2nd co-op: medical devices vs consumer electronics
- First time in California
- Graduate School...
- Asking Questions
- Coffee Chats



Official Co-op Statistics

1,234

Number of compliance
trainings completed

567

Berries picked, included
but not limited to:
strawberries, raspberries,
blueberries, blackberries,
boysenberries & mulberries

0

Number of cats I adopted
from Steve

8

Number of cats I wanted to
adopt from Steve

Questions

THANK YOU!

JEEVAN YERRAJI
ANDREA RAMIREZ
SUSAN GONG
ESTHER CHUNG
STEVE YEE
MIKE TODD
ALAN YU
HOKE AIHARA
CHITO BARCELO
DEEPA ADINARAYANAN
NING LI
WILTON LEE
VISHAL NARAYAN
RAHUL RAJGARHIA
DOUG DULL

ZOLTAN PISKOR
ALEC TOMHAVE
BEN RICHTER
LIDIA FLORES
ANNALISE WULF
BERNICE CHEN
SHYAM KONDURI
NAMI ANIL KUMAR
EVAN COHEN
DANNY ELIAHU
TINA KARIMI
JOHN HOANG
JASON LARSON
BHUPENDER SINGH
JULIE LEE

SANTOSH SURESH
CHRIS ZHANG
BILL CHAPIN
CHRISTINE BANKS
MICHAEL LESLIE
JUSTIN EDDIE
JP ONFFROY
LARRY LEE
AKIRA UEDA
JACK DIBACHI
JENNIFER HORAN
EDDIE WALKER
TIMOTHY REPPERT
GRACE RESTLE
MONTY LIONG

ANGELINA FARINAS
COURTNEY MCCRANE
NORALITIE LAUTI
TAYLOR FURTADO
JASON CHIN
RACHEL KWON
CHARLIE MEJIA
CIARA MANUEL
MEGAN TRAN
JUDY KIM

FELLOW INTERNS!
Everyone else who has
helped guide me along
the way!